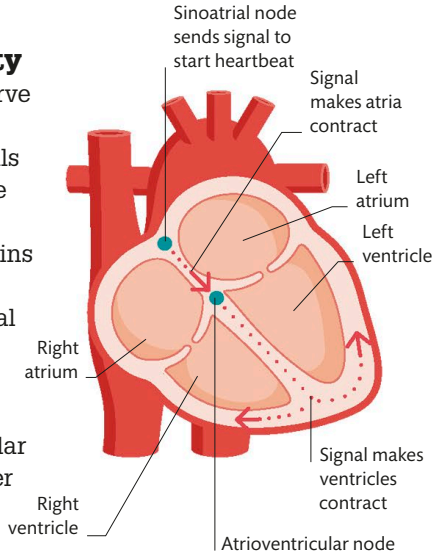
Pacemakers

A pacemaker is a battery-powered device implanted in the chest that corrects heartbeat abnormalities by sending electrical impulses to the heart.

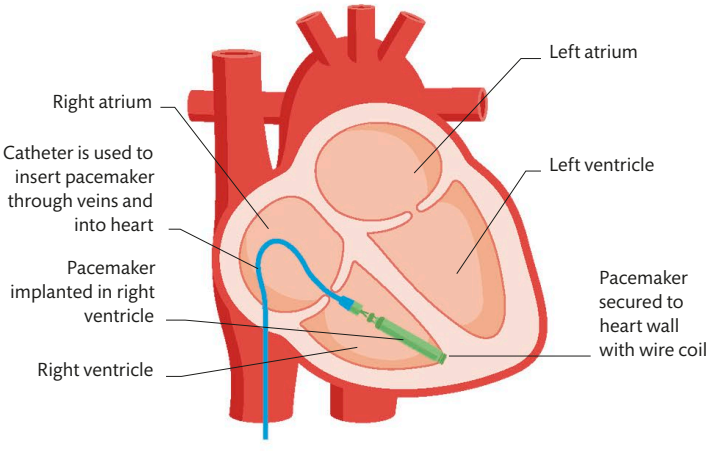
Normal heart activity

Heartbeats occur when nerve signals make the heart muscle contract. The signals come from patches of nerve tissue in the heart called nodes. Each heartbeat begins with a signal from the sinoatrial node, the "natural pacemaker," to make the upper chambers (atria) contract. The signal then passes to the atrioventricular node and down to the lower chambers (ventricles), making them contract.



Leadless pacemakers

Some pacemakers no longer need wires in order to work. These tiny devices are implanted directly into the right ventricle of the heart using a catheter. They contain a battery and a microchip that senses and, if necessary, corrects the heart rhythm. The microchip also transmits data to electrodes on the skin, enabling heart activity to be monitored externally.



CAN I USE A CELL PHONE IF I HAVE A PACEMAKER?

Yes, but the phone should be kept at least 6 in (15 cm) away from the pacemaker. There is no evidence that Wi-Fi or other wireless Internet devices interfere with pacemakers.

How pacemakers work

In some heart disorders, the heart's nodes do not work properly, so the heart beats too slowly, too fast, or with an abnormal rhythm. A pacemaker may be implanted into the patient's chest, to take over the role of the nodes and regulate the heartbeat. Some pacemakers act on one chamber of the heart, while others act on two or three chambers to ensure that the chambers work in a normal rhythm.

Biventricular pacemaker

This device is used for people with disorders, such as heart failure, in which the ventricles fail to contract at the same time. The pacemaker has three leads and sends signals to the right atrium and to both ventricles at once to synchronize contractions of the chambers. Treatment with a biventricular pacemaker is also sometimes called cardiac resynchronization therapy (CRT).

